Satellite Image Time-Series Data Augmentation Using an Attention Mechanism Variational Recurrent Autoencoder

Ferdaous Chaabane
COSIM Laboratory
Carthage University, SUPCOM
Tunis, Tunisia
0000-0001-6681-753X

Florence Tupin
Laboratoire Traitement et
Communication de l'Information
Telecom ParisTech
Paris, France
0000-0002-3110-8183

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***Abstract*— Data scarcity presents a significant challenge in satellite image analysis, particularly for developing robust models in remote sensing applications. High-quality and abundant data are essential for accurate predictions; however, acquiring Satellite Image Time-Series (SITS) data is often constrained by factors such as limited temporal coverage and the high cost of Very High Resolution (VHR) acquisitions. To address this issue, we propose a novel Attention-based Variational Recurrent Autoencoder (AVRAE) designed for generating synthetic satellite image time-series data. This method extends the evidence lower bound (ELBO) of variational inference to incorporate the temporal dependencies essential for satellite data. A recurrent neural network-based autoencoder framework is employed, integrated with an attention mechanism to effectively capture both short- and long-term temporal relationships. The AVRAE framework synthesizes realistic and statistically representative satellite time-series data, enabling enhanced analysis for remote sensing applications. Evaluations using real-world satellite datasets demonstrate that AVRAE produces coherent and statistically valid synthetic data, thereby improving VHR SITS data quality for deep learning-based remote sensing applications.**

Keywords— Satellite Image Time Series, Variational Recurrent Autoencoder, Attention, data generation, etc.

I. INTRODUCTION

Leveraging advanced generative techniques [1,2], synthetic satellite image time series can be generated with realistic temporal and spatial characteristics, enabling diverse applications in remote sensing, environmental monitoring, and disaster management. Numerous machine learning and deep learning techniques have been developed for synthetic data generation, spanning various fields, including SITS.

Generative Adversarial Networks (GANs) [3] use a generator and discriminator in a competitive framework to model data distributions. These networks automatically generate robust and representative features through hierarchical structures. However, traditional convolutional GANs often struggle to capture geometric or structural patterns consistent across certain classes. Self-attention mechanisms, by contrast, achieve a better balance between modeling long-range dependencies and computational efficiency. The self-attention module computes the response at a position as a weighted sum of features from all positions, making it particularly effective in classifying remote sensing time series [4]. Building on this foundation, the SAGAN version introduced in [5] incorporates long-range contextual details and multi-level dependencies across classes. Recently, SAGAN has been employed for hyperspectral data generation, demonstrating its versatility.

Graph Convolutional Neural Networks (GCNNs) have also been applied to generative tasks involving graph-structured data. Models like Spatio-Temporal Graph Convolutional Networks (STGCN) [6] extend these capabilities to satellite imagery by modeling relationships between extracted geographical regions and their temporal evolution [7].

Finally, Variational Autoencoders (VAEs) [8] combine the variational method with traditional autoencoders, enabling the learning of probabilistic latent representations. Extensions like VRAE models integrate RNNs with VAEs to handle sequential data, while models such as the Variational Transformer Network (VTN) [9] employ attention mechanisms for structured data generation. These techniques have shown promise for satellite image time series by encoding temporal and spatial features into latent representations to generate realistic synthetic sequences [10].

This paper introduces the attention-based variational recurrent autoencoder (AVRAE) for generating satellite image time-series data. Inspired by the variational recurrent autoencoder (VRAE) introduced in [11] for industrial control system data, AVRAE extends the evidence lower bound (ELBO) of variational inference to handle sequential data effectively. While VRAE maps sequential data to a latent space, it struggles with capturing the complex dynamics inherent to time-series data.

The innovative version of AVRAE proposed in this paper incorporates attention mechanisms, drawing from the transformer's scaled-dot attention. By calculating attention weights based on the encoder's and decoder's hidden states, the model identifies the relative importance of input sequences. These weights enhance the reconstruction and generation of sequences, enabling AVRAE to produce synthetic data that closely resemble real data both visually and statistically.

The paper is organized as follows: After reviewing the existing methods for synthetic data generation, the second section presents the flowchart of the proposed approach followed by the problem formulation through equations. Next section presents the benchmark VHR SITS dataset and explains that the results will be included in the last version of the paper.

II. METHODOLOGY FOR SITS DATA GENERATION

A. *flowchart*

Fig. 1 introduces an attention-based variational recurrent autoencoder (ARVAE) designed specifically for generating Satellite Image Time-Series data. ARVAE comprises an RNN-based encoder and decoder, tailored for processing sequential data. While RNNs can receive input at timestep t and produce corresponding output data in real-time, previous studies have highlighted the limitations of such a structure when applied to predictive models for sequential data. To address these shortcomings, we adopted an encoder-decoder framework. This architecture generates a context that encapsulates the entire input sequence and allows for outputs of variable lengths. A variational approach [12] is used to learn the probability distribution of the latent vector (the context) summarizing the input sequence. While this approach can develop robust generative models for time-series data, our preliminary experiments revealed its limitations in capturing the complexities of SITS data.

Indeed, SITS data consist of both discrete and continuous variables, with some exhibiting periodic short-term patterns and others characterized by long-term dependencies. To overcome this challenge, we enhanced the VRAE framework by incorporating an attention mechanism, enabling the model to better capture both temporal dynamics. Fig. 2 illustrates the RNN based variational Recurrent Autoencoder architecture.

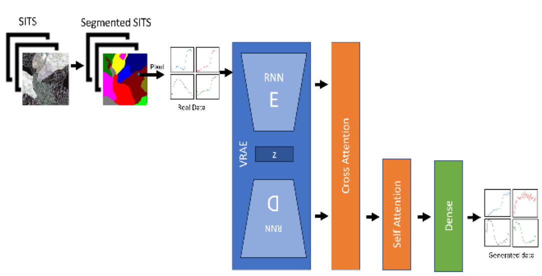

Fig. 1. AVRAE flowchart.

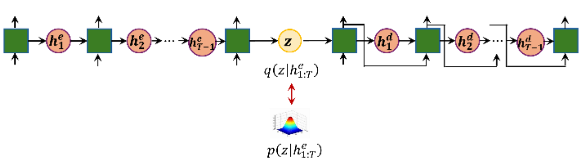

Fig. 2. Detailed Encoder/ Docoder architecture.

B. *Problem formulation*

In this section, we introduce the notation used throughout this work and address the use of AVRAE for the generation of high-quality multispectral time series samples.

We consider X as an input time series profile, $X = (x_0, x_1, \dots, x_T)$ of pixels $x_t \in R^D$ of the D multispectral bands.

The original multispectral times series profiles are transformed into a latent space using the AVRAE.

B.1. Inference process

The variational inference of VAE approximates the true posterior probability distribution p(z | x), where z represents latent variables and x represents observations, with an empirical distribution $q(z)$. The Kullback–Leibler (KL) divergence between these distributions is expressed as:

$$KL(q(z) \,||\, p(z \mid x)) = E_{q(z)}[\log q(z)] - E_{q(z)}[\log p(z,x)] + \log p(x) \quad (1)$$

Since directly computing p(z | x) is often infeasible, variational inference instead maximizes the Evidence Lower Bound ($ELBO$), derived as follows:

$$\log p(x) \geq E_{q(z)}[\log p(z,x)] - E_{q(z)}[\log q(z)] = ELBO(q) \quad (2)$$

In this formulation, maximizing the $ELBO$ minimizes the KL divergence, ensuring an optimal approximation of $q(z)$ to $p(z \mid x)$.

However, the basic $ELBO$ does not incorporate sequential dependencies within time-series data. To address this, we extend the $ELBO$ to sequential data by introducing latent states $h_{1:T}$ for observations $x_{1:T}$:

$$ELBO(q) = Eq_{(z,h_{1:T}|x_{1:T})}[\log p(z, h_{1:T}, x_{1:T})] - Eq_{(z,h_{1:T}|x_{1:T})}[\log q(z, h_{1:T} \mid x_{1:T})] \quad (3)$$

Using Bayes' rule [8] and the chain rule, the extended $ELBO$ can be broken into three components:

1) Reconstruction term: $Eq[log\, p(x_{1:T} \mid z)]$,
2) KL divergence for z: $-\mathrm{KL}(\mathrm{q}(\mathrm{z} \mid h_{1:T}) \| \mathrm{p}(\mathrm{z} | h_{1:T}))$,
3) KL divergence for $h_{1:T}$: $-\mathrm{KL}(\mathrm{q}(h_{1:T} \mid x_{1:T}) \| \mathrm{p}(h_{1:T}))$.

To maximize the ELBO, the model must:

- Minimize the KL divergences for z and $h_{1:T}$
- Maximize the reconstruction term by improving the generative process.

This optimization aligns with the encoder-decoder architecture, where the encoder learns to approximate latent variables, and the decoder reconstructs observations from these latent representations.

ARVAE combines the encoder-decoder framework with variational inference (cf. Fig 2). Two key adaptations distinguish its architecture:

- The hidden state h_t at each timestep follows a stochastic process.
- The latent variable z, representing the context, depends on the final hidden state of the encoder.

a) Hidden State Sampling

We assume that the hidden state h_t at the timestep t depends only on the hidden state of the previous timestep and the observation of that timestep. The hidden state at timestep t is sampled as:

$$h_t \sim q_\theta(h_t \mid h_{t-1}, x_t) \quad (4)$$

Where q_θ adopted in this paper is a Gaussian distribution parameterized by θ:

$$h_t \sim \mathrm{N}\left(\mu_{h_t}, \mathrm{diag}\left({\sigma_{h_t}}^2\right)\right), \quad [\mu_{h_t}, \sigma_{h_t}] = \phi(h_{t-1}, x_t) \quad (5)$$

b) Latent Variable Sampling

Similarly, the latent variable z is sampled at the final timestep T of the encoder:

$$z \sim q_{\mathrm{w}}(h_t) \quad (6)$$

where q_{w} is also a Gaussian distribution parameterized by w:

$$z \sim N\left(\mu_z, diag(\sigma_z{}^2)\right),\ [\mu_z, \sigma_z] = \phi(h_T) \quad (7)$$

The decoder initializes its hidden state with z and uses previous outputs as inputs to generate predictions. This generative approach ensures alignment with the reconstruction objective.

B.2. Attention Layer

Numerous studies have established that the attention mechanism significantly enhances the predictive performance of models designed for time-series data. This impact is particularly pronounced in encoder-decoder architectures. In traditional seq2seq models, a well-known example of the encoder-decoder framework, the decoder generates its output based solely on a fixed-size context vector produced by the encoder. However, this approach often struggles to effectively transmit all the input information to the decoder, especially as the distance between the first input and the last output grows. To improve sequence modeling, ARVAE employs both cross-attention and self-attention mechanisms introduced in [13].

a) Cross-Attention

Cross-attention captures the relationship between the encoder's input and decoder's output sequences, guiding the decoder to focus on relevant parts of the input. This is achieved by using the hidden states of both the encoder and decoder to compute attention weights as:

$$W_c = \mathrm{softmax}\left(\frac{H_{dec}H_{enc}{}^T}{\sqrt{d}}\right) \quad (8)$$

The final cross-attention output is:

$$O_c = \mathrm{LN}(\mathrm{MLP}(W_c H_{enc}) + H_{enc}) \quad (9)$$

b) Self-Attention

Self-attention refines the cross-attention output and is given by:

$$\begin{aligned} W_s &= \mathrm{softmax}\left(\frac{O_c O_c{}^T}{\sqrt{d}}\right) \\ O_s &= \mathrm{LN}(\mathrm{MLP}(W_s O_s) + O_c) \end{aligned} \quad (10)$$

B.3. Learning Process

The ARVAE learning process combines all components in a unified encoder-decoder architecture. The encoder processes sequential data to produce latent representations z and hidden states $h_{1:T}$. The decoder reconstructs the input sequence while optimizing ELBO to balance reconstruction accuracy and latent space regularization.

The inference steps for the encoder and decoder are described first, followed by their integration into the learning algorithm.

For the feedforward of the encoder, a minibatch $x_{1:T}$ of size m is processed as input and we obtain as outputs: H_{enc}, Λ, z, and $[\mu_z, \sigma_z]$, where Λ represents the parameters of the Gaussian distribution for the hidden states.

Besides, the decoder should be trained as a generative model. AVRAE decoder takes the hidden states H_{enc} and the latent variable z generated by the encoder as inputs. It reconstructs $\hat{x}_{1:T}$, initializing its hidden state with z. Rather than relying on actual observations as inputs, the decoder utilizes its own previously generated output $x_{d:t-1}$ from the recurrent operations at each timestep. Like the encoder, the decoder iteratively applies a recurrent operation ϕ_d to generate hidden states H_{dec} at each timestep t. It then computes O_s by sequentially applying cross-attention and self-attention mechanisms, leveraging the hidden states from both the encoder Henc and the decoder H_{dec}. Finally, the reconstructed sequence $\hat{x}_{1:T}$ is produced using O_s via a dense layer.

B.4. Generation Process

Traditional Variational Autoencoders (VAEs) primarily learn the latent variable distribution produced by the encoder. In such models, the decoder alone can generate synthetic data based on this distribution. However, AVRAE introduces additional complexity by working with sequential data and employing two latent variables. This makes it challenging to generate data solely using the decoder. Specifically, even if the hidden state is trained to follow a standard Gaussian distribution, its dependence on previous observations and states complicates the sampling process. To address these challenges, AVRAE uses both the encoder and decoder for data generation. This approach enables AVRAE to generate synthetic data that maintains structural similarity to the pixel sequence while introducing variability.

III. EXPERIMENTAL RESULTS

A. Dataset description

The available segmented and labeled multispectral time series data is derived from Sentinel-2 images covering Zaghouan region in Tunisia, spanning the period from January 2021 to January 2024.

To generate time series from Sentinel-2 pixels, the gray level (visible bands), NIR, and SWIR bands are considered because they provide complementary information about surface conditions. The visible bands help identify general surface changes and land cover types. The NIR band is highly sensitive to vegetation, making it useful for monitoring biomass and vegetation dynamics. SWIR bands are less affected by atmospheric conditions and are particularly effective for detecting soil and vegetation moisture, drought stress, and even thermal anomalies. Together, these spectral ranges allow for robust and interpretable time series, especially in environmental monitoring applications such as water detection, vegetation health, and land surface changes.

The performance of AVRAE is evaluated using a benchmark dataset composed of multiple pixels extracted from urban areas, forest regions, grasslands, and water bodies. Two key metrics have been used: the perceptual quality and the KL divergence. The first metric will help assess how closely the generated outputs resemble the original ones. The KL divergence will measure the difference between the learned latent distribution of the model and the predefined prior distribution. A lower KL divergence indicates that the latent space is well-structured and aligns effectively with the prior, which is essential for generating meaningful and diverse temporal pixel variations.

B. Results

The methodology proposed in this work has been implemented using an adaptation version of the source code provided by [11] to multispectral times series data. The time series of the selected pixels (real data) have been filtered to reduce the noise caused by clouds' appearance.

Figure 3 shows the KL divergence over the epochs during the training of the ARVAE generative model. It displays a decreasing trend, which indicates that the model gradually learn to align its latent distribution with the prior. The KL divergence goes under 1.0 when reaching 100 epochs.

Fig. 3. KL divergence over the epochs.

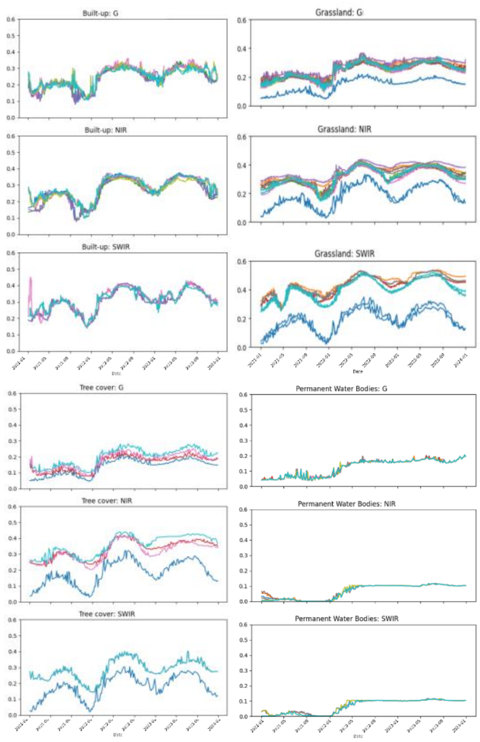

Fig. 4. A visual comparision between real data (blue curve) and several generated curves (other colors) for the urban areas, forest regions, grasslands, and water bodies. The bands Gray, NIR and SWIR are considered.

The blue lines represent the real data. Five synthetic data were generated for each single pixel according to ARVAE model. Overall, synthetic time series are assembled in a similar pattern near the real data. The temporal patterns of the generated curves closely match those of the real data, showing similar dynamics. However, some differences in amplitude are observed, which can be attributed to the selection of pixels from different zones to prepare the benchmark dataset for the training.

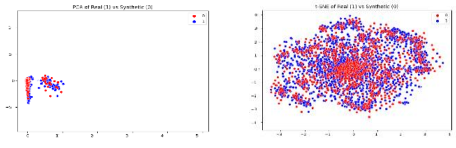

Fig. 5. Visualization for PCA (left) and t-SNE (right).

Besides, the plausibility of the synthetic data was further confirmed through visualizations with PCA and t-SNE in Figure 5. PCA preserved data variance, while t-SNE emphasized the local structure and cluster separation in the high-dimensional space. In both visualizations, the synthetic data (red) showed similar distribution and variance to the real data (blue), suggesting that both datasets lie close on the same manifold. These results indicate that the synthetic data is difficult to distinguish from the real data in the reduced-dimensional space.

IV. Conclusion

In this study, we proposed a generative signal model based on a Variational Autoencoder architecture enhanced with an attention mechanism, aiming to improve both latent space alignment and signal reconstruction quality. Selected metrics demonstrated the model's capacity to generate signals that preserve meaningful structural and temporal characteristics. These results validate the relevance of integrating attention mechanisms in variational frameworks, especially for time-dependent or structured data like signals. Future work will explore more expressive priors and multi-modal inputs to further enhance generation diversity and fidelity.

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